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To the Editor of [journal name — to be chosen by the authors],

**Submission: "Choosing a Fast Stand-In for a Slow Simulation: a Selection Tool and a 44-Problem Study"**

**The practical problem.** Engineers who rely on a slow simulation can afford only a limited number of runs, and then have to decide which fast stand-in model to train and whether more runs would be worth the cost. Published comparisons of Gaussian-process, reduced-basis and neural-network stand-ins disagree, are often run under unequal tuning, and do not give a procedure a practitioner can run on their own data.

**What the paper offers.** (1) An open tool, SurroSelect, that chooses a stand-in by cross-validation, reports which models are tied, checks whether the POD basis limits accuracy, and forecasts the error for more runs. (2) A pre-specified evaluation of the tool (criteria committed to version control before the evaluation runs) on eight problems solved by an independent finite-element library, against a tuned neural network, and on two real measured datasets. (3) A study of 44 problems describing where each family wins, with controls for the reduced basis, the regressor and the width of the hyper-parameter search.

**The evidence, in numbers.** The criteria were first tested on eight problems; the tool was then frozen, with predictions written down, and tested on eight further ones. On the further problems selection regret against the oracle had median 1.04 (95% interval 1.00–1.11), and the tuned network was better than the tool by more than 25% in two of 16 cells. Several results went against the tool or against our predictions: it did not match the benchmark's own selection among methods that include networks (in either batch), the learning-curve forecast was within a factor two in only 68% of 96 cases over all sixteen problems, the basis check raised no flag on the further problems, and the benchmark finding that a reduced-basis model wins at large budgets held on six of eight of them where we had predicted that it would not. Adding a tuned network to the pool helped on one of two wave problems, at several times the cost, and on a solver taking 1.52 seconds per run the surrogate was cheaper than the solver after about 335 predictions. Limitations are listed in the paper: the problems are simulated and their solvers are fast, the real-data check uses scalar noisy outputs, the networks were tuned with a limited budget, and the forecast is a rough extrapolation.

**Why this journal.** [To be written by the authors after choosing the journal and checking its scope, article types, indexing and author guidelines; the project record does not contain a verified match for the authors' constraints.]

**Statements.** The manuscript is original and is not under review elsewhere [authors to confirm]. All data are generated by the code in the project repository or are public, openly licensed measurements (CC BY 4.0); the tool is released under the MIT licence; a permanent archive with a DOI [to be provided]. The study design, code, analyses and drafting used an AI system (Claude) under the authors' direction, as disclosed in the acknowledgments; the internal reviews were AI-agent reviews, not human peer review; all authors must approve the submission.

Sincerely,
[Author names, affiliations, corresponding-author e-mail — to be supplied by the authors]